

C Programming Mini Project Report Template

(2nd Year Engineering Students)

Project Information

Field	Details
Course Name	C Programming Fundamentals
Project Title	Student Result Management System
Project Type	Console-Based Application
Programming Language	C
Development Environment	Code::Blocks / Dev C++ / Visual Studio Code
Version Control	Git & GitHub
Team Size	2–4 Students
Project Duration	2–3 Weeks
Difficulty Level	Beginner to Intermediate

1. Problem Statement

Section	Description
Existing Problem	Educational institutions often maintain student records manually, making it difficult to search, update, calculate results, and generate reports efficiently.
Proposed Solution	Develop a console-based Student Result Management System using C Programming to automate student record management.

Target Users	Faculty Members, Lab Assistants, Academic Coordinators
Expected Benefits	Faster record management, improved accuracy, automatic result calculation, reduced manual effort, and better organization of student data.

2. Project Objectives

Objective ID	Objective
OBJ-01	Add new student records
OBJ-02	Store subject-wise marks
OBJ-03	Calculate Total Marks
OBJ-04	Calculate Percentage
OBJ-05	Generate Student Grade
OBJ-06	Search Student Records
OBJ-07	Update Student Details
OBJ-08	Delete Student Records
OBJ-09	Store and Retrieve Data using File Handling

3. User Stories

User Story ID	User Story	Acceptance Criteria	Priority
US-01	As an Admin, I want to add a new student so that student information can be maintained.	Student details are saved successfully and duplicate Roll Numbers are not allowed.	High
US-02	As an Admin, I want to search a student using Roll Number so that records can be retrieved quickly.	Correct student details are displayed when a valid Roll Number is entered.	High

US-03	As an Admin, I want to update student marks so that records remain accurate.	Updated marks are saved successfully.	High
US-04	As an Admin, I want to delete student records so that obsolete records are removed.	Selected student records are permanently deleted.	Medium
US-05	As an Admin, I want the system to calculate percentage and grade automatically.	Percentage and grade are calculated correctly based on marks entered.	High
US-06	As an Admin, I want to store student records in a file so that data is available even after restarting the application.	Student records are successfully saved and loaded using file handling.	High

4. Functional Requirements

Requirement ID	Requirement	Status
FR-01	Add Student	✓
FR-02	Display Student Details	✓
FR-03	Search Student	✓
FR-04	Update Student Details	✓
FR-05	Delete Student	✓
FR-06	Calculate Result	✓
FR-07	Generate Grade	✓
FR-08	Save Records to File	✓
FR-09	Load Records from File	✓

5. Non-Functional Requirements

Category	Requirement
Performance	Response time should be less than 2 seconds.
Reliability	Accurate result calculation with proper validation.
Usability	Simple menu-driven interface for easy navigation.
Maintainability	Modular programming using functions.
Portability	Program should execute on Windows and Linux systems.
Security	Input validation for Roll Number and Marks.

6. Project Modules

Module	Description	C Concepts Used
Student Registration	Add student information	Structures, Functions
Marks Entry	Enter subject-wise marks	Arrays
Result Calculation	Calculate Total, Percentage and Grade	Functions
Student Search	Search records using Roll Number	Loops
Update Record	Modify existing student details	Functions
Delete Record	Remove student records	Arrays
File Management	Save and retrieve records	File Handling

7. C Programming Concepts Used

Concept	Implementation
Variables	Store marks, totals, percentages

Data Types	int, float, char
Arrays	Store multiple student records
Functions	Modular program development
Structures	Student Record Management
Loops	Menu-driven operations
Switch Case	Menu selection
Conditional Statements	Input validation and decision making
File Handling	Permanent storage of records

8. Expected Program Features

Feature	Description
Add Student	Register new students
Display Students	View all student records
Search Student	Find student by Roll Number
Update Student	Modify student details
Delete Student	Remove student record
Calculate Result	Generate Total and Percentage
Generate Grade	Display grade automatically
Save Data	Store records in text file
Exit	Close application safely

9. Test Cases

Test Case ID	Scenario	Input	Expected Output	Status
TC-01	Add Student	Valid Student Details	Student Added Successfully	Pass
TC-02	Duplicate Roll Number	Existing Roll Number	Duplicate Record Error	Pass
TC-03	Search Student	Valid Roll Number	Student Details Displayed	Pass
TC-04	Search Student	Invalid Roll Number	Student Not Found	Pass
TC-05	Update Student	Updated Marks	Record Updated Successfully	Pass
TC-06	Delete Student	Valid Roll Number	Student Deleted Successfully	Pass
TC-07	Save File	Student Data	File Saved Successfully	Pass
TC-08	Load File	Existing File	Records Loaded Successfully	Pass

10. Expected Deliverables

Deliverable	Description
Source Code	Complete C Project
Project Report	PDF Report
PowerPoint Presentation	Project Presentation
GitHub Repository	Public Repository
README File	Project Documentation
Output Screenshots	Minimum 5 Program Screenshots
Test Cases	Test Execution Report

11. GitHub Repository Structure

Student-Result-Management-System

```
— SourceCode
  |— main.c
  |— student.c
  |— student.h
— Report
  |— Project_Report.pdf
— PPT
  |— Presentation.pptx
— Output
  |— Screenshots
— Data
  |— student.txt
— README.md
— LICENSE
```

12. GitHub Submission Guidelines

Item	Requirement
Repository Visibility	Public
Repository Name	Student-Result-Management-System-C
README File	Mandatory
Source Code	Complete and Executable
Report	PDF Format

Presentation	PPT/PDF Format
Screenshots	Minimum 5
Commit History	Minimum 10 Meaningful Commits
Team Members	Mention in README
License	Optional (MIT Recommended)

Sample Git Commit Messages

- Initial Project Setup
 - Added Student Registration Module
 - Implemented Search Functionality
 - Added File Handling
 - Updated Grade Calculation Logic
 - Fixed Search Bug
 - Improved Menu Interface
 - Added Screenshots
 - Updated README
 - Final Project Submission
-

13. GitHub Evaluation Rubric (50 Marks)

Evaluation Criteria	Marks
Repository Structure	5
Code Organization	10
Meaningful Commit History	5
README Documentation	10
Code Quality	10
Screenshots & Output	5
Project Report Upload	5
Total	50

14. PowerPoint Presentation Guidelines

Slide No.	Content
1	Title Slide (Project Title, Team Members, Guide)
2	Problem Statement
3	Objectives
4	User Stories
5	Functional Requirements
6	System Flowchart
7	Project Modules
8	C Programming Concepts Used
9	Code Explanation
10	Program Demonstration
11	Output Screenshots
12	Challenges Faced
13	Future Enhancements
14	GitHub Repository Demo
15	Conclusion & Thank You

15. Presentation Evaluation Rubric (80 Marks)

Evaluation Criteria	Marks
---------------------	-------

Problem Statement	5
Requirement Analysis	5
User Stories	5
System Design	10
Code Explanation	15
Program Demonstration	10
Communication Skills	10
PPT Design & Content	10
Time Management	5
Question Handling	5
Total	80

16. Viva Evaluation Rubric (100 Marks)

Evaluation Criteria	Marks
Understanding of Problem Statement	10
Requirement Analysis	10
C Programming Fundamentals	15
Functions, Arrays & Structures	10
File Handling Concepts	10
Logic & Algorithm Explanation	10
Code Walkthrough	10
Program Demonstration	10
GitHub Repository Explanation	5

Communication & Technical Confidence	10
Total	100

17. Sample Viva Questions

Question No.	Question	Expected Outcome
1	Explain the problem statement of your project.	Requirement Understanding
2	Why did you choose Structures?	Data Organization Knowledge
3	Why are Functions important in C?	Modular Programming
4	Explain File Handling used in your project.	Data Persistence
5	What is the role of Arrays?	Data Storage
6	Explain the difference between while and do-while loops.	Loop Concepts
7	How is Percentage calculated?	Program Logic
8	How did you test your application?	Testing Knowledge
9	Explain your GitHub repository structure.	Version Control Knowledge
10	What future enhancements can be added to this project?	Software Design Thinking

18. Expected Learning Outcomes

Outcome ID	Learning Outcome
LO-01	Apply problem-solving techniques to build a real-world application using C.

LO-02	Design modular programs using functions and structures.
LO-03	Implement arrays, loops, conditional statements, and file handling effectively.
LO-04	Develop debugging and testing skills using sample test cases.
LO-05	Use Git and GitHub for source code management and collaboration.
LO-06	Prepare professional project documentation and presentations.
LO-07	Demonstrate technical concepts confidently during project presentations and viva examinations.

Submission Checklist

Item	Status (✓/✗)
Source Code Completed	
Code Compiles Successfully	
Test Cases Executed	
README Prepared	
GitHub Repository Created	
Minimum 10 Commits Completed	
Project Report Submitted	
PPT Prepared	
Output Screenshots Added	
Viva Preparation Completed	

Document Version: 1.0

Course: C Programming Fundamentals

Recommended Audience: 2nd Year Engineering. Students

Purpose: Mini Project Report Template for Academic Evaluation and Project Presentation